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Beyond RMSE: Do the most accurate building load forecasts deliver the most operational value?

ByungKi Jeon, Min-Hwi Kim *, Jong-Kyu Kim

Renewable Energy System Laboratory, Korea Institute of Energy Research (KIER), 152 Gajeong-ro, Yuseong-gu, Daejeon 34129, Republic of Korea

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ABSTRACT

Short-term building load forecasting is almost universally evaluated with statistical error metrics—root-mean-square error (RMSE), mean absolute error (MAE) and the coefficient of determination (R^2). Such forecasts are nonetheless used for operating decisions: battery scheduling, peak shaving, pre-cooling. We examine whether the lowest-RMSE model yields the best operating outcomes. Using two years of hourly data from 200 buildings (12 sites, 16 use types) in the Building Data Genome Project 2, we train six day-ahead forecasters (seasonal-naïve, ridge regression, random forest, gradient boosting, a multilayer perceptron and an LSTM) and evaluate each on operational value: a forecast-driven battery dispatch (time-of-use arbitrage and demand-charge peak shaving), peak forecasting skill, and an illustrative pre-cooling case. The RMSE-best model is the value-best model in only 12% of buildings—no better than chance (17.2% under an empirical independence null; permutation $p = 0.047$). The within-building rank correlation between RMSE and economic value is just 0.17 and is non-positive in 32% of buildings. Selecting by RMSE costs a median of $181 and a mean of $2872 per building per year under the primary tariff (the mean is concentrated in the largest buildings), rising under a demand-charge-heavy tariff. A key mechanism is the training loss: symmetric error minimization regresses peaks toward the conditional central tendency, under-forecasting the demand-charge-setting peaks by 5–11% of the annual peak. Retraining the same gradient-boosting model with an asymmetric pinball loss ($\tau = 0.9$) reverses the peak bias and raises mean captured value by 84% despite a 19% worse RMSE. Peak-coincident error is the strongest—though still weak—accuracy correlate of value ($\rho = 0.274$ versus 0.170 for RMSE). The findings are robust to tariff structure, battery sizing, oracle horizon, battery degradation, weather-forecast noise, random seeds and hyperparameters, and persist at the site-portfolio level. We recommend that benchmarks report value-based metrics alongside accuracy for peak-sensitive applications, and provide a framework for doing so.

1. Introduction

Data-driven short-term load forecasting is a mature and active research area. New models are proposed continuously and are almost always justified by a reduction in a statistical error metric—typically RMSE, MAE, mean absolute percentage error (MAPE) or CV-RMSE [1–4]—relative to a set of baselines [5]. This evaluation paradigm implicitly assumes that lower error is uniformly better: that a model which is more accurate on average will also perform better in every downstream use of the forecast.

In building operation, however, forecasts are inputs to decisions, and the value of a forecast is the quality of the decisions it enables rather than the size of its average error [6,7]. A behind-the-meter battery scheduled for peak shaving depends almost entirely on the few hours that set the monthly demand charge; a day-ahead pre-cooling controller depends on correctly anticipating the afternoon ramp; a demand-response dispatcher depends on peak timing. These objectives are not average-error quantities. A forecast can have a low RMSE while systematically targeting the wrong hour, or a high RMSE while accurately anticipating the peaks. Average accuracy and decision value can therefore, in principle, diverge; where they do, an evaluation based on average error may not be aligned with operational objectives.

Whether they diverge, and by how much, is an empirical question that has not been addressed at scale for buildings. Most studies that examine the value of forecasting consider a single building or a single season, a single application, or compare only two or three models, leaving open whether any divergence is an idiosyncrasy or a general phenomenon. This study addresses that gap. Building-level value is also distinct from, and less studied than, system-level value in wholesale-

* Corresponding author.
E-mail address: mhkim001@kier.re.kr (M.-H. Kim).

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Nomenclature

$A_{d,h}$, $P_{d,h}$	actual / predicted load on day d, hour h, (kWh)
$\widehat{A}$	annual peak of the actual load, kW
$\overline{a}$	annual mean of the actual load, kW
f_h	forecast building load in hour h, (kWh)
c_h, d_h	battery charge / discharge power in hour h, (kWh)
SoC_h	battery state of charge at hour h, (kWh)
π_h	time-of-use energy price in hour h, $/kWh
λ	demand-charge rate, $/kW-month
$P_{\max}$	battery power rating, kW
E_{cap}	battery energy capacity, kWh
P_{run}	realized running monthly peak demand, kW
η_c, η_d	charge / discharge efficiency ($\eta_c = \eta_d = \sqrt{0.90}$), –
e	monthly peak-slack (epigraph) variable, kW
b_{nobatt}, b_{model}, b_{oracle}	annual bill: no battery / forecast-driven / perfect-forecast, $
T_t, $T_{\mathrm{out},t}$	zone / outdoor air temperature in hour t, °C
C, R	zone thermal capacitance / resistance, kWh/°C, °C/kW
$Q_{c,t}$, Q_{cap}	delivered / available cooling power, kW
s_d	forecast on-peak severity of day d (0–1), –

Abbreviation

RMSE	root-mean-square error
MAE / MAPE / sMAPE	mean absolute (percentage / symmetric percentage) error
CV-RMSE	coefficient of variation of the RMSE
MPC	model predictive control
LP	linear program
TOU	time-of-use (energy tariff)
DC	demand charge
BESS	battery energy storage system
BDG2	Building Data Genome Project 2
LSTM / MLP / RF	long short-term memory / multilayer perceptron / random forest
HistGBM	histogram-based gradient-boosting machine
EFC	equivalent full cycle

market and unit-commitment settings, even though commercial building bills—dominated by time-of-use energy and monthly demand charges—are a setting in which a single building-level decision translates forecast quality directly into cost.

1.1. Contribution

- We formalize and quantify the gap between statistical accuracy and operational value for day-ahead building load forecasts, across 200 buildings spanning many climates and use types, over a full year of out-of-sample operation (avoiding season-specific artefacts).
- We build an operational-value framework that turns any forecast into realized outcomes: a closed-loop, forecast-driven battery MPC (TOU energy plus demand charge), peak-management metrics, and an illustrative pre-cooling case study.
- We show that the lowest-RMSE model is frequently not the highest-value model, quantify the economic regret of selecting by RMSE, and identify the mechanism (peak under-forecasting by symmetric-loss training)—which we verify causally by retraining the same model class under asymmetric losses.
- We show which accuracy metrics best proxy operational value, test whether value-based selection is implementable ex ante, and

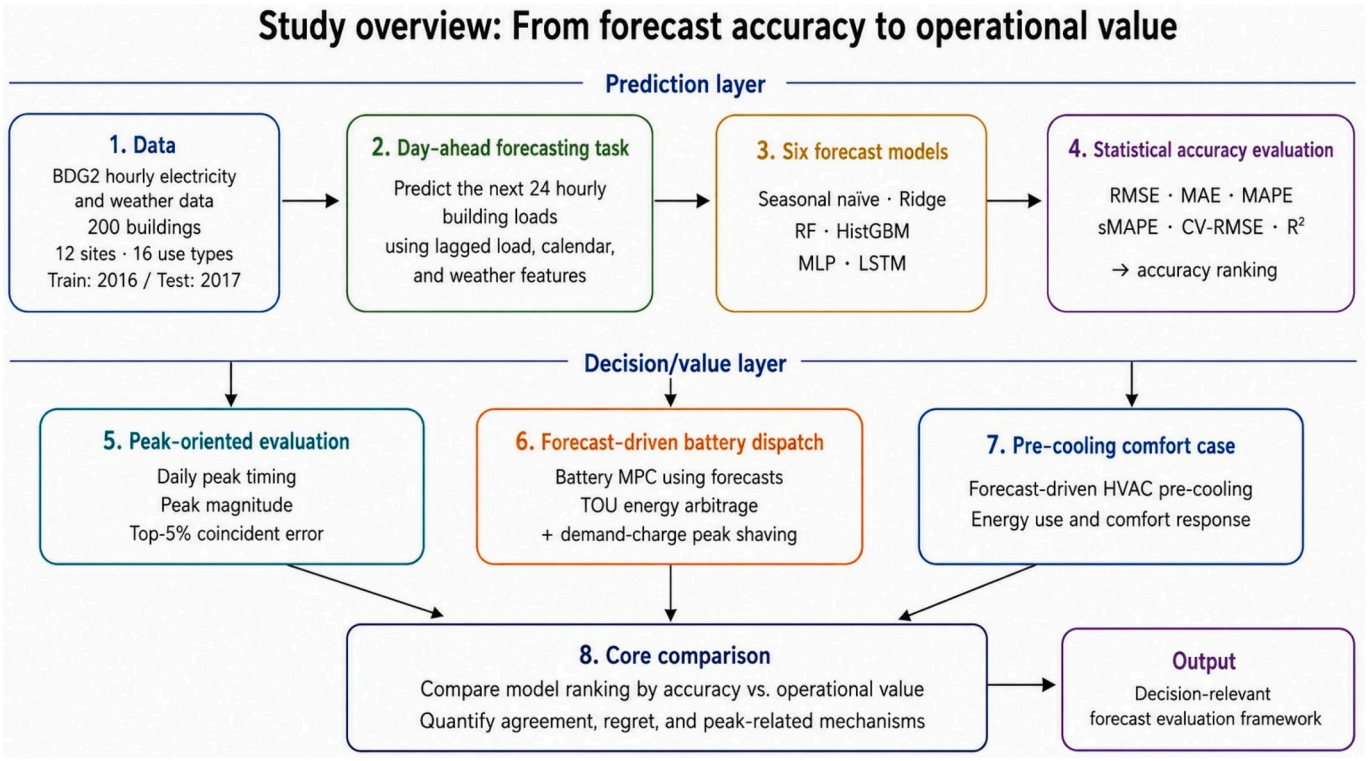


Fig. 1. Study overview: the prediction layer produces six day-ahead forecasts and their accuracy ranking; the decision/value layer converts the same forecasts into peak metrics, battery-dispatch economics and a pre-cooling case, whose value ranking is compared against the accuracy ranking.

recommend a decision-relevant evaluation protocol for peak-sensitive applications.

Fig. 1 summarizes the study. A prediction layer trains and scores the six day-ahead forecasters by conventional statistical accuracy, while a decision/value layer passes the same forecasts through peak-oriented evaluation, a forecast-driven battery model-predictive controller and a pre-cooling comfort case; the core comparison then contrasts the accuracy ranking with the operational-value ranking.

2. Background and related work

Forecast evaluation in buildings. Data-driven building load forecasting has matured rapidly, progressing from regression and tree ensembles to deep recurrent and attention-based architectures [8–11], and several reviews now survey the field and the machine-learning pipeline behind it [1–4,12–14]. The trend continues in this journal: recent Energy and Buildings studies contribute hybrid deep architectures [15], decomposition–Transformer pipelines for industrial buildings, multi-horizon deep learning benchmarks [16], spatio-temporal graph networks and Transformer transfer-learning studies trained on the same BDG2 dataset used here [17,18], temporal-fusion-transformer transfer learning [19], and large multi-building benchmarks [20]. Across this literature—including large-scale forecasting competitions [21–23] and measurement-and-verification guidance such as the CV-RMSE thresholds of ASHRAE Guideline 14 [24]—models are evaluated almost exclusively with statistical error metrics: RMSE, MAE, MAPE, CV-RMSE and R^2. Notably, a benchmark of eleven models on 56 non-domestic buildings found that model ranking depends more on building load dispersion than on algorithm sophistication [20], and a recent review in this journal explicitly frames the field's open challenge as moving "from prediction to decision" [25]. The broader forecasting literature has long studied what such scores do and do not capture, from proper scoring rules to comprehensive surveys of forecasting practice [5,26,27]. These metrics share three properties that matter here: they are averages over all hours, they are symmetric in the sign of the error, and they are decision-agnostic [6]. None of these properties is obviously appropriate when the forecast will be used to make a peak-sensitive, asymmetric, decision-specific choice.

Value of forecast accuracy. A smaller but growing literature in power systems studies the economic value of forecast accuracy rather than its statistical size [28]—for example the cost of wind and solar forecast errors in scheduling and trading [29], value-oriented forecasting that selects the forecast (or quantile) maximizing a downstream objective rather than minimizing error [7,30], and cost- or value-oriented load forecasting that targets the operating cost directly [31–33]. Closely related, probabilistic load forecasts trained with asymmetric (pinball) losses make the differing costs of over- and under-prediction explicit [34]. Recent storage-arbitrage studies report the same phenomenon we quantify here: statistically more accurate price forecasts do not necessarily raise storage profits, and tuning the training loss on downstream value does [35]. For buildings specifically, two recent Energy and Buildings studies are the closest antecedents: Li et al. [36] quantify the value of load-forecast information for storage MPC on a single-building case and find the economics highly sensitive to forecast error only when demand charges are present, and Langtry et al. [37] find that sizeable changes in load-forecast accuracy translate into surprisingly small changes in MPC operating cost. A field deployment likewise reported that the most accurate prediction model was not the most useful one in operation [38], and a systematic comparison of five forecasters feeding a storage MPC found the accuracy ranking and the control-performance ranking to disagree [39]. These studies consider one or a few buildings or a single controller; what has been missing—and what this paper contributes—is a population-scale contrast of the ranking of many models by accuracy against their ranking by realized value, together with the mechanism that drives the divergence.

Decision-focused and value-based learning. A parallel line in machine learning—"decision-focused learning", "smart predict-then-optimize" and "predictive-to-prescriptive" analytics—argues that models should be trained and judged on downstream decision cost rather than prediction error [40–44], and this lens has recently reached energy applications such as storage arbitrage, renewable-energy trading and PV-battery scheduling [45–47]. A recent multi-building application is instructive: a decision-focused LSTM more than doubled its RMSE relative to an accuracy-trained baseline yet reduced electricity costs across twenty buildings [47]. Our study is complementary and deliberately narrower: we keep the models trained conventionally and bring the decision-cost lens to evaluation at scale, so as to isolate and measure the accuracy-versus-value gap that motivates decision-focused training in the first place.

Behind-the-meter storage, demand charges and forecast-driven control. Commercial electricity bills are dominated by time-of-use energy and monthly demand charges—which in the United States are frequently 30–70% of the bill [48]—and behind-the-meter batteries dispatched by model-predictive control reduce both through energy arbitrage and peak shaving [49,50]. Model-predictive control is now a mature paradigm for buildings, supported by comprehensive tutorials, data-driven variants and field demonstrations [51–53], including recent work on handling forecast uncertainty in DER control [54] and field-tested forecast-driven pre-cooling [55,56], while reinforcement learning has emerged as a data-driven alternative for HVAC and whole-building control and for demand response [57–60]. The performance of all of these controllers depends on the forecast that drives them: the realized savings depend on correctly anticipating the peak-coincident hours, which is the mechanism by which forecast quality is converted into measurable operational value in this study.

Data efficiency and transferability. Because a per-building model must often be trained from limited history, transfer learning has been proposed to reuse knowledge across buildings and tasks [18,19,61–64], and pretrained time-series foundation models now promise zero- and few-shot forecasting without per-building training [65,66]. These directions change how a forecast is produced but not how it should be judged, and best-practice and open-benchmark efforts in energy forecasting [67,68] are naturally complementary to the value-based protocol we develop here.

3. Data and study design

We use the Building Data Genome Project 2 (BDG2) [69], an open collection of 3053 hourly energy meters from 1636 non-residential buildings at sites on several continents, spanning 2016–2017, with matched site weather. Of the 1578 buildings with electricity meters, we retain those satisfying three eligibility rules evaluated per building over 2016–2017: (i) at least 90% of hours observed in each of the two years; (ii) a strictly positive mean and standard deviation; and (iii) fewer than 5% non-positive readings. This yields 1052 candidate buildings. From these we draw a stratified random sample of 200 buildings using a fixed seed (NumPy default_rng(0)): buildings are allocated to the 16 primary-use categories proportionally to their frequency among the candidates, subject to a floor of three buildings per use type (or all available buildings if a category has fewer than three), with the totals then adjusted—largest categories first—until exactly 200 buildings are selected; within each category, buildings are drawn uniformly at random. The resulting allocation (Education 75, Office 36, Entertainment/public assembly 23, Public services 22, Lodging/residential 13, and eleven smaller use types with 2–3 buildings each) spans 12 sites, latitudes 30.3°–53.3° N, floor areas from about 1000 to over 860,000 ft^2, and per-building mean loads from under 1 kW to over 1500 kW (median $\approx$ 64 kW). The complete list of the 200 building identifiers, with use type and site, is given in Appendix A (and as a machine-readable table in the Supplementary Material), which together with the fixed seed makes the

sample exactly reproducible.

Temporal protocol. For every building we train on calendar year 2016 and evaluate on the entire 2017 test year (8760 h). Reporting a full year—rather than a representative week—removes seasonal selection bias and lets us break results down by season. Gaps of up to 24 consecutive hours in the load series are filled by time interpolation (longer gaps are forward/backward filled so that a continuous series exists for the dispatch simulation), and accuracy is scored only on originally observed hours. Site weather series are interpolated for gaps up to 12 h and median-filled thereafter.

4. Methodology

4.1. Day-ahead forecasting task

We target the day-ahead problem: at the start of each day the forecaster produces the 24 hourly loads of that day, using only information available by the end of the previous day. This is the horizon relevant to battery scheduling, demand-charge management and day-ahead demand response. All predictors respect this constraint: load lags are at least 24 h, previous-day and previous-week aggregates use fully observed days, and weather at the target hour is treated as a perfect day-ahead forecast—a common assumption in day-ahead studies. Because building peaks are weather-driven, realistic weather-forecast error would perturb every model alike and, if anything, widen the accuracy–value gap, so this case is conservative for our conclusion; we confirm in §5.8 that injecting realistic weather-forecast noise leaves the gap unchanged. Table 1 lists the complete 22-dimensional feature set.

4.2. Forecasting models

We deliberately span a wide accuracy range rather than seek a single winner, so that the accuracy-versus-value relationship can be traced across models of differing skill. Table 2 summarizes the six models and their principal hyperparameters; the complete training configuration (losses, optimizers, learning-rate schedules, early stopping, validation splits and software versions) is given in Supplementary Section S2.

All models are trained per building (the realistic deployment) with fixed random seeds, and every preprocessing statistic—feature standardization for Ridge and the MLP, and the load/temperature normalization of the LSTM—is fitted on the 2016 training year only. Negative predictions are clipped at zero. Tabular models produce each hour of the target day directly from that hour's feature row (direct multi-step formulation); the LSTM emits all 24 h jointly. Sensitivity of the headline results to the random seeds, to alternative hyperparameters and to the composition of the model set is reported in §5.8.

4.3. Accuracy metrics

On the test year we compute RMSE, MAE, MAPE, sMAPE, CV-RMSE and R^2 on observed hours. These constitute the conventional evaluation against which operational value is contrasted.

Table 1
The 22 input features (all available at the day-ahead decision time).

Group	Features (count)
Calendar	hour of day, day of week, month, weekend indicator (4)
Cyclical encodings	sin/cos of hour, sin/cos of day-of-week, sin/cos of day-of-year (6)
Weather at target hour	air temperature, dew-point temperature, wind speed, cloud coverage, daily-mean air temperature (5)
Load lags ($\geq$ 24 h)	load at $t-24$ h, $t-48$ h, $t-168$ h (3)
Daily aggregates	previous-day mean/max/min load, previous-week (day −7) mean load (4)

Table 2
Models and principal hyperparameters (scikit-learn 1.x / PyTorch 2.x defaults elsewhere; random seed 0 throughout).

Model	Configuration	Loss / training
Seasonal-naïve	load repeats from the same hour one week earlier (lag-168)	none (no fitted parameters)
Ridge	$\alpha = 1.0$ on standardized features	squared error (closed form)
RF	300 trees, unrestricted depth	variance (squared-error) split criterion
HistGBM	500 iterations, learning rate 0.05, L2 = 1.0 (scikit-learn HistGradientBoostingRegressor, algorithmically equivalent to LightGBM [70])	squared error
MLP	layers (128, 64), ReLU, L2 $\alpha = 10^{-3}$, Adam, $\leq$400 epochs, early stopping (10% validation split, patience 15)	squared error (MSE)
LSTM	seq2seq day-ahead network [71]: encoder LSTM (hidden 64) over the previous 168 h of normalized load, standardized temperature and cyclical calendar signals; head maps the encoding plus the target day's known exogenous features (24 temperatures, day-of-week one-hot, weekend flag) to 24 hourly outputs; Adam (lr $3{\cdot}10^{-3}$, weight decay 10^{-5}), batch 32, $\leq$160 epochs, ReduceLROnPlateau, gradient clipping at 1.0, last 12% of training days for validation, early stopping (patience 20, best weights restored)	MSE on load normalized by the training-year mean

4.4. Operational value I — peak management

Peak management is the first decision context: demand charges are levied on the monthly peak, demand-response commitments hinge on when the peak occurs, and pre-cooling must lead the afternoon ramp. We therefore compute six decision-relevant peak metrics directly from each forecast. Let $A_{d,h}$ and $P_{d,h}$ be the actual and predicted load on day $d \in \{1, \ldots, D\}$ and hour $h \in \{0, \ldots, 23\}$ of the test year, $\widehat{A}$ the annual peak of the actual load, $\bar{a}$ its annual mean, and $\mathscr{D}$ the set of days whose actual daily peak exceeds 20% of the median daily peak (excluding near-idle days such as holidays in unoccupied buildings).

Daily peak-timing error (h)—how far off the model places the hour of the daily maximum, the quantity a demand-response dispatcher needs:

$$\mathrm{PTE} = \frac{1}{D}\sum_{d=1}^{D}\left|\operatorname*{argmax}_h A_{d,h} - \operatorname*{argmax}_h P_{d,h}\right|$$

Daily peak-magnitude error (% of annual peak)—how well the model predicts the size of the daily maximum, which determines how much storage or curtailment must be positioned:

$$\mathrm{PME} = \frac{100}{|\mathscr{D}|\,\widehat{A}}\sum_{d\in\mathscr{D}}\left|\max_h P_{d,h} - \max_h A_{d,h}\right|$$

Signed daily peak bias (% of annual peak; negative ⇒ peaks are under-forecast)—the directional counterpart of (2), which diagnoses the systematic attenuation analysed in §5.5:

$$\mathrm{PB} = \frac{100}{|\mathscr{D}|\,\widehat{A}}\sum_{d\in\mathscr{D}}\left(\max_h P_{d,h} - \max_h A_{d,h}\right)$$

Monthly-peak error (% of annual peak)—the error in the quantity that is actually billed under a demand charge:

$$\mathrm{MPE} = \frac{100}{12\,\widehat{A}}\sum_{m=1}^{12}\left|\max_{t\in m} P_t - \max_{t\in m} A_t\right|$$

Top-5% coincident CV-RMSE (%)—the forecast error measured only over the hours in which the realized load is in its top 5% ($q_{0.95}$ denotes the 95th percentile of the hourly actual load). These are the hours that set demand charges and stress downstream decisions; a model can have a low all-hours CV-RMSE while being poor exactly here:

$$\mathrm{CV-RMSE}_{\mathrm{peak}} = \frac{100}{\bar{a}} \sqrt{\frac{1}{|\mathscr{T}_{95}|} \sum_{t \in \mathscr{T}_{95}} (P_t - A_t)^2}, \quad \mathscr{T}_{95} = \{t : A_t \geq q_{0.95}(A)\}$$

Morning-ramp onset timing error (h)—for cooling-dominated days, the error in when the load first crosses half of its daily range, searched between 04:00 and 14:00; this is the signal a pre-cooling controller must anticipate. With $o(X_d)$ the first hour $h \in [4, 14]$ at which $X_{d,h} \geq \min_h X_{d,h} + 0.5\left(\max_h X_{d,h} - \min_h X_{d,h}\right)$, computed over days where the onset exists for both series:

$$\mathrm{ROE} = \mathrm{mean}_d |o(A_d) - o(P_d)|$$

These six metrics were selected to cover, respectively, the *when* (1, 6), the *how much* (2,3) and the *billed outcome* (4) of peak behaviour, plus the error concentrated in the decision-relevant hours (5). We note one important asymmetry of definition: metric (5) is an ex-post diagnostic—it conditions on the hours that the realized load identifies as the top 5%, so it is a scoring tool for offline evaluation rather than a quantity available to a controller in real time. In §5.10 we show that a trailing-window version of it (computed on the preceding six months) can nonetheless be used ex ante for model selection.

4.5. Operational value II — forecast-driven battery dispatch

Scenario and controller. The second decision context is the workhorse of behind-the-meter economics: a battery that performs TOU energy arbitrage and demand-charge peak shaving. We size the battery per building at 25% of the 2016 peak load in power ($P_{\max}$) and two hours in energy ($E_{\mathrm{cap}} = 2P_{\max}$), with 90% round-trip efficiency ($\eta_c = \eta_d = \sqrt{0.90}$) and a 10–100% state-of-charge window. Each day, a model-predictive controller receives the day-ahead forecast f_h ($h = 1, \ldots, 24$) and solves the linear program

$$\begin{aligned} \min_{c,d,\mathrm{SoC},e} \quad & \sum_{h=1}^{24} \pi_h (c_h - d_h) + \lambda e \\ \text{s.t.} \quad & \mathrm{SoC}_h = \mathrm{SoC}_{h-1} + \eta_c c_h - d_h/\eta_d, \qquad \mathrm{SoC}_0 \text{ given}, \\ & 0 \leq c_h \leq P_{\max}, \quad 0 \leq d_h \leq P_{\max}, \quad \mathrm{SoC}_{\min} \leq \mathrm{SoC}_h \leq \mathrm{SoC}_{\max}, \\ & d_h - c_h \leq f_h \quad (\text{no export}), \\ & f_h + c_h - d_h \leq P_{\mathrm{run}} + e, \quad e \geq 0, \end{aligned}$$

where π_h is the TOU energy price in hour h (\$/kWh), λ the demand-charge rate (\$/kW-month), c_h and d_h the battery charge and discharge power (kW), SoC_h the state of charge (kWh), P_{run} the realized running peak of the current billing month so far (kW, reset at month boundaries), and e an epigraph slack variable that prices any increase of the monthly peak beyond P_{run} at the demand rate λ. The first term of the objective is the day's energy cost of battery operation; the second term is the marginal demand-charge exposure. The no-export constraint prevents discharging beyond the forecast building load.

Closed-loop application and feasibility. The optimized plan (c_h, d_h) is then applied to the *actual* load: discharge is curtailed if it would exceed the actual load plus concurrent charge (no export against the realized load), and charge/discharge are curtailed if they would violate the SoC bounds. The realized state of charge and the realized running peak P_{run} are carried forward to the next day; the end-of-day SoC is not forced back to any set level, and the simulation runs continuously over the whole test year. The realized bill (TOU energy plus twelve monthly demand charges) is accumulated from the realized net load. Because forecast quality is the only difference between models, differences in realized bills are attributable to the forecasts. In the LP solutions, planned simultaneous charging and discharging arises only as an objective-neutral end-of-day degeneracy in 0.045% of hours (and never in the terminal-SoC or degradation-aware variants of §5.7); it is retained by the feasibility layer but has no measurable effect on bills.

References and the capture rate. For each building we compute three reference quantities: The no-battery bill b_{nobatt}; the *perfect-forecast benchmark* b_{oracle}, obtained by running the identical receding 24 h controller with the actual load in place of the forecast, which isolates forecast quality under the same information-timing and controller; and each model's realized bill b_{model}. The headline value metric is the capture rate

$$\mathrm{capture} = \frac{b_{\mathrm{nobatt}} - b_{\mathrm{model}}}{b_{\mathrm{nobatt}} - b_{\mathrm{oracle}}},$$

the fraction of the perfect-forecast savings that a forecast actually captures (1 = perfect-forecast benchmark, 0 = no value). We use the capture rate as the headline value metric because it normalizes away building size and tariff scale—two buildings whose achievable savings differ by an order of magnitude become comparable—so that value can be aggregated and correlated across a heterogeneous population; the corresponding dollar quantities are reported alongside it wherever the economic magnitude matters.

Because the daily controller re-optimizes each day within a month-long billing objective, the receding-horizon perfect-forecast benchmark is not the global optimum over the billing period. We therefore also compute, for every building, a *monthly perfect-foresight optimum*: a single LP per billing month (the analogue of (7) over all hours of the month, with one monthly peak variable and the SoC chained across months). Under the primary tariff the receding daily benchmark attains 85.5% of the monthly optimum's savings in aggregate (per-building median gap 5.1% of achievable savings), and §5.7 shows that re-basing the capture denominator to the monthly optimum leaves every within-building ranking—and hence every agreement, correlation and regret statistic—exactly unchanged, because the denominator is constant within a building. A 48 h receding perfect-foresight variant closes about a quarter of the daily-versus-monthly gap.

Properties of the capture definition. Two properties warrant note. The denominator is the achievable saving $(b_{\mathrm{nobatt}} - b_{\mathrm{oracle}})$, so the ratio becomes numerically sensitive for buildings whose achievable saving is small, and it is unbounded below: a forecast whose dispatch raises the bill above the no-battery baseline yields a capture below zero. Only 2% of the 200 buildings have achievable savings under \$200/yr, but negative captures do occur (in 22–31% of buildings depending on the model under the primary tariff; §5.3 reports the full distributions). To limit the influence of these tails we report medians alongside means for all dollar-valued quantities; moreover, because b_{nobatt} and b_{oracle} are identical across the competing models within a building, the within-building ranking of models by capture is invariant to the denominator, so the central accuracy-versus-value comparisons do not depend on it.

Tariffs. We use a primary commercial tariff (TOU energy with a weekday on-peak window of 16:00–21:00 at \$0.26/kWh, a mid-peak rate of \$0.16/kWh for 07:00–16:00 and 21:00–23:00, an off-peak and weekend rate of \$0.09/kWh, plus an \$18/kW-month demand charge) and report sensitivity to three further tariff archetypes (flat + DC; TOU-only; demand-heavy at \$30/kW with compressed energy prices) and to battery size in §5.6. Battery throughput, equivalent full cycles and degradation-cost sensitivity are reported in §5.7.

4.6. Operational value III — pre-cooling comfort case (secondary, illustrative)

The third decision context ties the same peak-forecast skill to HVAC operation. On the 181 buildings at warm-climate sites (summer outdoor maximum $\geq$32 °C), a representative conditioned zone—driven by each building's real site weather—participates in a daily on-peak demand-response event (16:00–21:00) during which HVAC cooling is curtailed; to ride through the event, the controller pre-cools the zone beforehand (12:00–16:00), with a depth scaled by the forecast of that day's on-peak severity. The zone is a lumped resistance–capacitance (RC) model with an hourly time step:

$$T_{t+1} = T_t + \frac{1}{C}\left[\frac{T_{\text{out},t} - T_t}{R} + Q_{\text{int},t} + Q_{\text{sol},t} - Q_{c,t}\right],$$

where T_t is the zone temperature, $C = 6$ kWh/°C the thermal capacitance, $R = 6$ °C/kW the resistance to outdoor, $Q_{\text{int},t}$ the internal gain (1.5 kW during occupied weekday hours 08:00–19:00, 0.4 kW otherwise), $Q_{\text{sol},t}$ a clear-sky solar gain (a daytime half-sine of amplitude 1.5 kW attenuated up to 60% by the site's observed cloud cover), and $Q_{c,t}$ the delivered cooling. Cooling follows the setpoint $T_{\text{set},t}$ subject to the available capacity $Q_{\text{cap},t}$:

$$Q_{c,t} = \min\left(Q_{\text{cap},t}, C\left[T_t^{\text{free}} - T_{\text{set},t}\right]^+\right),$$

where T_t^{free} is the temperature the zone would reach in hour t without cooling and $[x]^+ = \max(x, 0)$. The cooling capacity is $Q_{\max} = 12$ kW normally and $\gamma Q_{\max}$ during the on-peak event ($\gamma = 0$, full curtailment, in the base case). The controller's only forecast-dependent decision is the pre-cool depth: for each day d the forecast on-peak severity is

$$s_d = \text{clip}\left(\frac{\overline{F}_{\text{on},d} - L_{10}}{L_{90} - L_{10}}, 0, 1\right), \qquad T_{\text{set},t} = T_{\text{base}} - s_d\,\Delta_{\text{pre}} \text{ for } t \in 12:00\text{–}16:00,$$

where $\overline{F}_{\text{on},d}$ is the model's mean forecast load over that day's on-peak window, L_{10} and L_{90} are the 10th and 90th percentiles of the building's 2016 daily on-peak means (so severity is calibrated on training data only), $T_{\text{base}} = 24$ °C and $\Delta_{\text{pre}} = 5$ °C. A model that over-forecasts on-peak severity pre-cools deeper than necessary (wasting cooling energy); one that under-forecasts pre-cools too little (risking discomfort during the curtailment).

Two outcome metrics are reported over the cooling season (May–September, occupied weekday hours): *seasonal cooling energy* (kWh, of which pre-cooling energy is the 12:00–16:00 share) and *comfort-violation degree-hours* $\sum_t [T_t - 25^\circ\text{C}]^+$ (with violation hours counted above a 0.1 °C threshold). These two metrics are precisely the two failure modes that average-error metrics cannot separate: energy waste from over-forecasting and discomfort from under-forecasting. To test robustness of the qualitative mechanism we sweep eight zone configurations (base; thermal mass C = 4 and 9 kWh/°C; insulation $R = 4$ and 8 °C/kW; partial curtailment $\gamma = 0.2$; comfort band 26 °C; pre-cool depth 3 °C); quantitative results for all eight are given in §5.11 and Supplementary Section S6. The case is illustrative—a single lumped zone with a stylized controller does not generalize universally—but it isolates a forecast-driven HVAC cost that RMSE does not see.

4.7. Analysis

For each building we rank the six models by each accuracy metric and by each operational metric, and compute the within-building Spearman rank correlation between them; we report the distribution across buildings. We measure the selection agreement (how often the metric-best model is the value-best model) and the economic regret of choosing the operating model by RMSE: with m_b^{RMSE} the RMSE-best and m_b^* the capture-best model in building b, and $s_b(m)$ the realized annual bill saving of model m,

$$\text{regret}_b = s_b\left(m_b^*\right) - s_b\left(m_b^{\text{RMSE}}\right) \geq 0.$$

We report the regret distribution in dollars, in capture and as a share of the annual bill. Finally we rank the candidate accuracy and peak metrics by their within-building rank correlation with operational value, to identify the best operational proxies, and we quantify the peak-under-forecasting mechanism. For the two headline statistics we also report inferential checks that respect the small within-building sample of six models: the observed selection agreement is compared against an empirical independence null (computed from the marginal win distributions of the six models, with a permutation test across buildings), and the mean within-building Spearman correlation is tested against the exact null distribution of a six-item rank correlation (§5.3).

5. Results

5.1. The most accurate model is metric-dependent

Table 3 reports accuracy across the 200 buildings. The ranking is already metric-dependent: RMSE ranks the LSTM first (40.4), ahead of HistGBM (42.9) and Ridge (43.2), whereas MAE, CV-RMSE and R^2 all rank Ridge first (19.3, 34.9, 0.54), with the LSTM second. The MLP has a negative mean R^2 (−0.29): it performs poorly on a minority of buildings, which lowers its average despite a competitive typical-day error. Before any operational consideration, the identity of the best model is therefore not uniquely defined.

Because CV-RMSE is scale-free, its level also invites comparison with other studies. Two features of Table 3 matter for that comparison. First, the *mean* CV-RMSE (34.9–55.2%) is heavily inflated by a tail of small and irregular buildings: the *median* CV-RMSE is 15.3–21.6%, and the within-model correlation between CV-RMSE and building mean load is strongly negative (Spearman ≈ −0.5); the 16 buildings with mean load below 10 kW have a median CV-RMSE of 53% (Ridge) against 15% for the remainder. Second, this heavy upper tail is a documented property of heterogeneous non-residential populations rather than of any particular model: the error analysis of the ASHRAE Great Energy Predictor III competition—whose meters underlie BDG2—found that roughly 5% of test data could not be predicted acceptably by any of the top 50 solutions and a further 16% only with difficulty [72,73]; the BuildingsBench day-ahead benchmark reports aggregate NRMSE of about 13–17% for commercial buildings but 77–80% for residential ones across 1900+ buildings [68]; and a recent analysis of 611 BDG2 buildings shows that 70% of the cross-building variance in day-ahead CV-RMSE is explained by intrinsic load variability rather than model choice [74]. Our median levels are consistent with these benchmarks, and our per-building models use fixed, untuned hyperparameters (§4.2)—per-building tuning would lower absolute errors but, as §5.8 shows, does not change the accuracy-versus-value conclusion.

5.2. Operational value yields a different model ranking

Under the primary tariff the mean capture rate is HistGBM 0.140 ≈ RF 0.138 ≈ LSTM 0.130 ≈ Ridge 0.129 ≫ MLP 0.102 ≫ Naïve 0.054 (Table 4): the four mid-range models capture similar value, substantially more than the high-variance MLP and the seasonal-naïve baseline.

Fig. 2 makes the mechanism concrete for one building: the lowest-RMSE model (Ridge) smooths and under-shoots the peaks, so its battery dispatch leaves the demand-charge-setting monthly peaks higher, while a less accurate model (HistGBM) that tracks the peaks shaves them toward the perfect-forecast benchmark, capturing more than twice the value despite a higher RMSE.

The decoupling is general. The RMSE leader (LSTM) is mid-pack on value; HistGBM, second on RMSE, leads on value. The divergence is most pronounced in the win rates: Ridge wins on RMSE in 32.5% of buildings but on value in only 19.5%, while the MLP—worst on mean

Table 3
Accuracy across 200 buildings: mean (and median CV-RMSE); best mean in bold.

Model	RMSE	MAE	CV-RMSE (%)	median CV-RMSE (%)	R^2
Naïve	95.3	26.4	39.3	21.6	0.35
Ridge	43.2	**19.3**	**34.9**	15.3	**0.54**
RF	44.9	22.5	45.4	15.4	0.35
HistGBM	42.9	22.5	42.7	15.5	0.40
MLP	50.3	27.4	55.2	19.1	−0.29
LSTM	**40.4**	20.4	36.4	17.1	0.45

Table 4
Accuracy versus operational value, primary tariff (TOU + demand charge). "Win" = share of the 200 buildings where the model is best on that criterion.

Model	mean RMSE	mean capture	win on RMSE (%)	win on value (%)
Naïve	95.3	0.054	3.5	9.0
Ridge	43.2	0.129	32.5	19.5
RF	44.9	0.138	18.5	11.0
HistGBM	42.9	0.140	30.0	17.0
MLP	50.3	0.102	1.0	21.5
LSTM	40.4	0.130	14.5	22.0

RMSE and the RMSE winner in just 1.0% of buildings—is the value-best model in 21.5% of them, and the LSTM tops the value wins (22.0%). Plotted as rankings, the accuracy and value orders cross rather than coincide (Fig. 3); ordering the six models by accuracy shows the same at the distribution level (Fig. 4).

5.3. *RMSE rarely selects the value-best model, and the economic regret is measurable*

Across the 200 buildings the RMSE-best model is also the value-best model in only 12.0% of cases (Wilson 95% CI 8.2–17.2%). Because the six models have unequal win probabilities, the appropriate chance benchmark is not only the uniform 16.7% but the independence null computed from the empirical win distributions of Table 4, which is 17.2%; a permutation test across buildings (20,000 shuffles) places the observed 12.0% below this null (two-sided $p = 0.047$). Selecting by RMSE is therefore no better than selecting at random—nominally, it is slightly worse. No accuracy metric exceeds 14% (MAE 13.5%, MAPE 13.0%, sMAPE 12.5%). This near-chance top-one agreement does not contradict the weak positive rank correlation reported next: because the six models have tightly clustered realized values (Table 4), a weak accuracy-versus-value association cannot reliably identify the single most valuable model.

The within-building Spearman correlation between RMSE skill and capture is 0.17 (median 0.17) and is non-positive in 32% of buildings—in roughly one building in three, more accuracy is associated with less value. With only six models per building, an individual within-building correlation is a noisy quantity (the exact null standard deviation of a six-item Spearman coefficient is 0.447); the appropriate inference is on the population: the mean of 0.170 over 200 buildings is 5.4 null standard errors above zero (permutation $p < 5{\cdot}10^{-5}$), and 68% of buildings have a positive coefficient. The association is therefore reliably positive but small. It also depends heavily on the weakest model: excluding the seasonal-naïve baseline—poor on both accuracy and value—the mean within-building correlation among the five trained models falls to 0.04, i.e., among realistic candidate models, accuracy is essentially uninformative about value. (Excluding any single model, the top-one agreement remains at or below the five-model chance level of 20%; §5.8.) Across all model pairs, the more accurate model is the more valuable one only 57% of the time, marginally above the 50% chance level (Fig. 5).

Selecting models by RMSE is value-suboptimal in 88% of buildings, costing, under the primary tariff, a median of $181 per building per year and a mean of $2872 (equivalently 0.57 percentage points of bill savings; the quartiles are $39–$547, the 90th percentile $2407, the 95th $3980 and the 99th $20,847). The mean is dominated by the largest buildings: the top decile of buildings carries 89.6% of the total regret, the 5%-trimmed mean is $479, and excluding the 5% largest buildings (by 2016 peak) the mean is $548; regret correlates with building size (Spearman 0.50 with the 2016 peak) and is largest for commercial uses with pronounced peaks (Fig. 6). For context, the achievable (perfect-forecast) savings themselves are a median of $6101 and a mean of $18,266 per building-year (5.2% and 6.2% of the annual bill, respectively), and only 2% of buildings have achievable savings under $200. Negative captures—dispatch that raises the bill above the no-battery baseline—occur for 22–31% of buildings depending on the model, a further reminder that forecast-driven dispatch value is asymmetric and tail-driven in a way that average accuracy metrics do not describe.

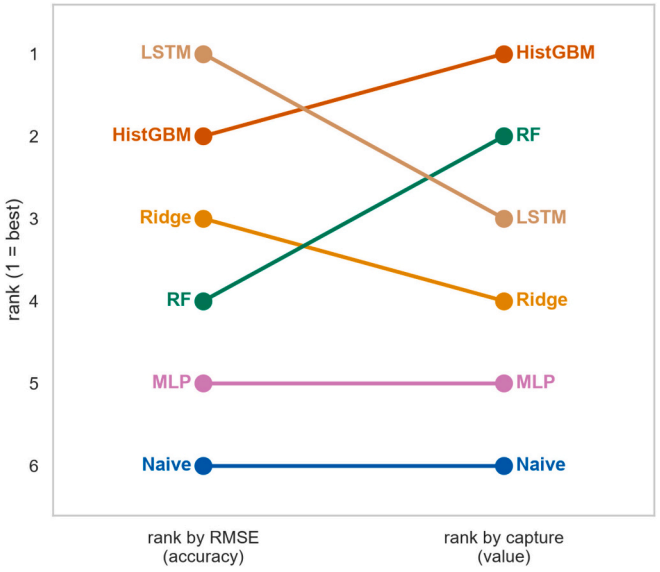


Fig. 3. Model rank by accuracy (RMSE) versus by operational value (capture): the two orders cross rather than coincide.

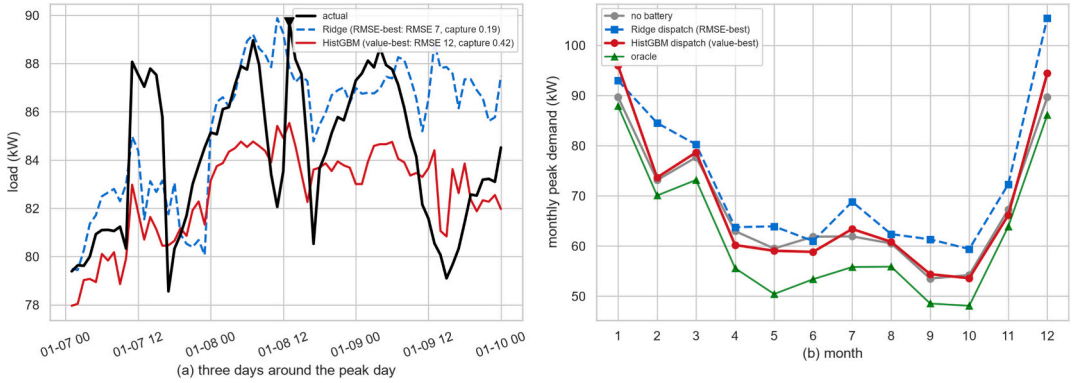


Fig. 2. Mechanism in one building (Rat_education_Jacob). (a) Around the peak day, the RMSE-best model (Ridge) under-shoots the peaks that the value-best model (HistGBM) tracks. (b) The resulting monthly peaks after dispatch: the value-best forecast shaves them toward the perfect-forecast benchmark.

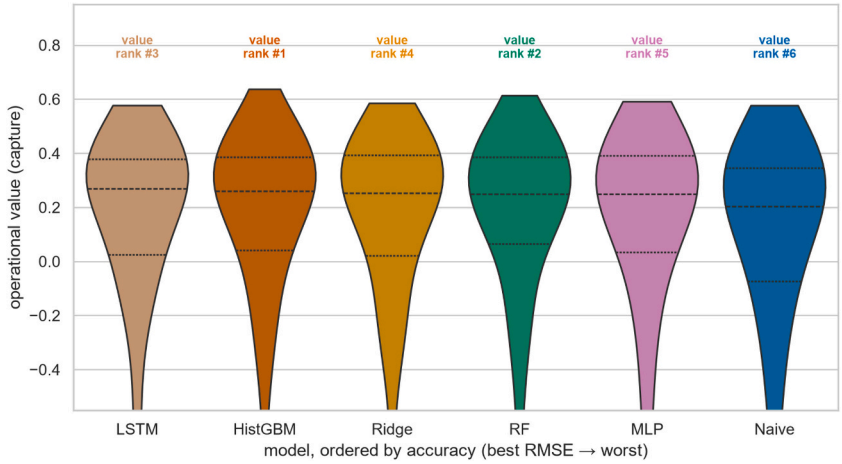


Fig. 4. Capture distributions across the 200 buildings with models ordered by mean RMSE (best first); the value ranks (labels) do not follow the accuracy order.

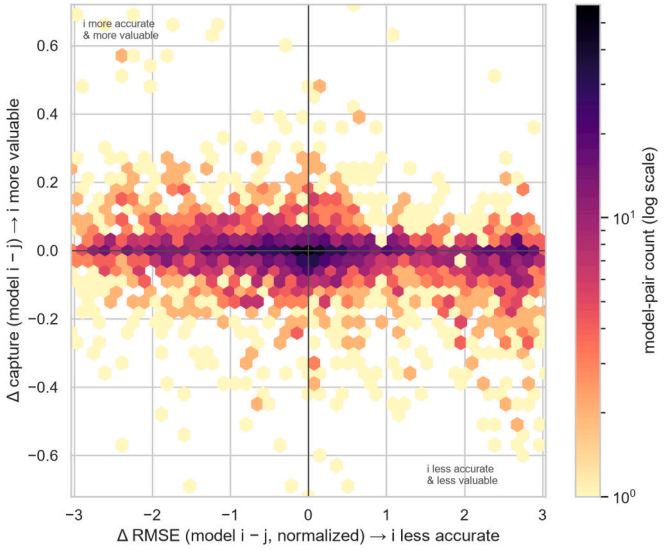


Fig. 5. Pairwise differences in RMSE versus in capture, all model pairs in all buildings: the more accurate model is the more valuable one in only 57% of pairs.

5.4. *Peak-coincident error is a stronger correlate of value than RMSE*

Ranking every candidate metric by its mean within-building correlation with operational value, the strongest correlates are peak-coincident CV-RMSE (mean Spearman $\rho = 0.274$) and peak-magnitude error ($\rho = 0.225$), both exceeding RMSE ($\rho = 0.170$); the peak-coincident correlation is about 1.6 times the RMSE correlation (Table 5). We emphasize that all of these correlations are weak in

Table 5
Mean within-building Spearman rank correlation ρ between each metric (as skill) and operational value (capture), with 95% confidence intervals across the 200 buildings, sorted.

Metric	mean ρ	95% CI	Type
Peak-coincident CV-RMSE	0.274	[0.210, 0.338]	peak
Peak-magnitude error	0.225	[0.164, 0.287]	peak
sMAPE	0.172	[0.112, 0.232]	accuracy
RMSE	0.170	[0.110, 0.230]	accuracy
R^2	0.170	[0.110, 0.230]	accuracy
CV-RMSE	0.170	[0.110, 0.230]	accuracy
MAE	0.146	[0.083, 0.210]	accuracy
MAPE	0.144	[0.081, 0.207]	accuracy
Ramp-onset timing	0.141	[0.076, 0.207]	peak
Peak-time error	0.132	[0.066, 0.198]	peak
Monthly-peak error	−0.084	[−0.152, −0.016]	peak
Absolute peak bias	−0.088	[−0.156, −0.019]	peak

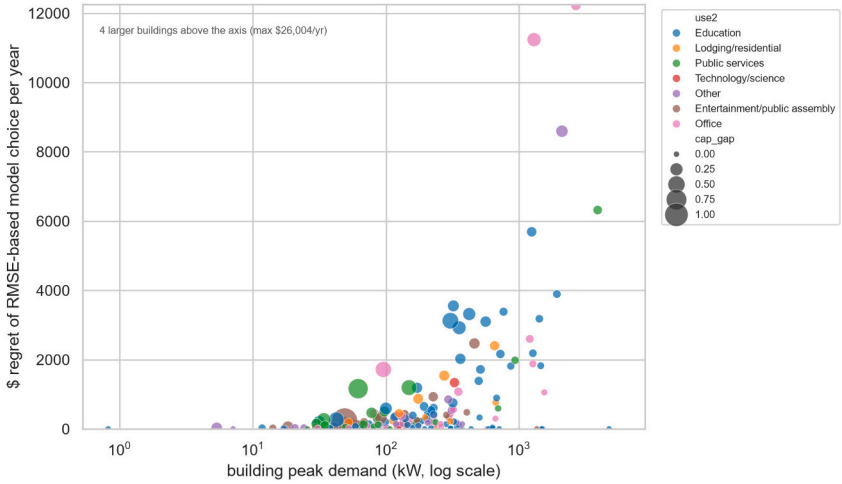


Fig. 6. Annual regret of RMSE-based model choice versus building peak demand (log axis); bubble size is the capture gap, colour the use type.

absolute terms: the conventional average-error metrics cluster tightly at $\rho \approx 0.14$–0.17, so the claim is not that any single metric strongly predicts value, but that the peak-focused metrics are consistently—if modestly—more informative than the average-error metrics. That is, the error during the top-5% demand hours—the hours that set the demand charge—is the single best statistical proxy for the operational value of the forecast. A clustered map of the metric–metric rank correlations shows the same structure: operational value clusters with the peak metrics, separate from the tightly inter-correlated RMSE/MAE/CV-RMSE/sMAPE/R^2 group (Fig. 7).

5.5. Mechanism: Symmetric-loss training under-forecasts peaks—and changing the loss reverses it

Conditioning the peak forecast bias on the size of each day's peak illustrates the mechanism (Fig. 8): every model over-forecasts the small daily peaks but under-forecasts the largest ones—by 5–11% of the annual peak at the top percentiles—because minimizing a symmetric average error pulls predictions toward the conditional central tendency and attenuates the extremes. Averaged over all days this leaves a net negative peak bias for every model trained with a symmetric loss (Ridge −2.1%, RF −1.7%, HistGBM −1.4%, LSTM −0.9% of the annual peak); the untrained seasonal-naïve baseline is nearly unbiased (−0.4%), and only the MLP over-forecasts (+1.3%), reflecting optimization noise under early stopping rather than its loss. Because peak-shaving value depends on positioning storage for the true peak, this systematic under-forecasting reduces operational value—and explains both why a more accurate model can yield lower value and why the over-forecasting MLP is the value-best model in roughly one-fifth of buildings despite its poor accuracy.

The conditional-mean argument, however, is only a hypothesis until the loss is isolated from model capacity, smoothing and hyper-parameters. We therefore retrained the *same* model class (HistGBM), with identical features, data, hyperparameters and seed, changing *only* the training loss, on all 200 buildings (Fig. 9). The result is unambiguous. With the squared-error loss the mean peak bias is −1.4% and the mean capture 0.140 (all four variants were retrained under the revision software environment, so the squared-error baseline differs marginally from the archived Table 3 run—RMSE 43.4 versus 42.9; the within-experiment comparison is exact). Switching to the absolute (L1) loss—still symmetric, now targeting the conditional median—makes the peak bias *more* negative (−2.5%, as expected for right-skewed load distributions) even though it attains the best RMSE of the four variants; capture rises only marginally (0.151). Switching to an asymmetric pinball loss shifts the bias toward zero ($\tau = 0.7$: −0.5%) and past it ($\tau = 0.9$: +2.9%), progressively reduces the peak-coincident error (64.7 → 56.1%), and raises the mean capture to 0.191 ($\tau = 0.7$) and 0.258 ($\tau = 0.9$)—an 84% increase over the squared-error configuration—while the RMSE *worsens* from 43.4 to 51.7. The $\tau = 0.9$ variant captures more value than its squared-error twin in 84% of the 200 buildings, with a median gain of $351 per building-year. Since everything except the loss is held fixed, this establishes the training loss as a causal driver of the systematic peak attenuation—one that operates independently of model capacity and data—and shows that the accuracy-versus-value divergence can be created, and reversed, at the loss function. The controlled manipulation covers one model class; the consistent negative peak bias of every symmetric-loss model in Fig. 8 is corroborating—though observational—evidence that the same mechanism operates across classes.

5.6. Robustness I: Tariff structure and battery size

The gap is robust and is largest where peak charges contribute most to the bill (Table 6). As the demand charge grows in importance, the agreement between RMSE and value falls (16.9% → 11.5%), the mean regret of an RMSE-driven choice rises to $5505 per building per year under the demand-heavy tariff (median $343)—this figure is the average across buildings in that scenario, not the cost at a worst-case building—and peak-coincident error overtakes the average-error metrics as the best value predictor. Under a pure time-of-use tariff with no

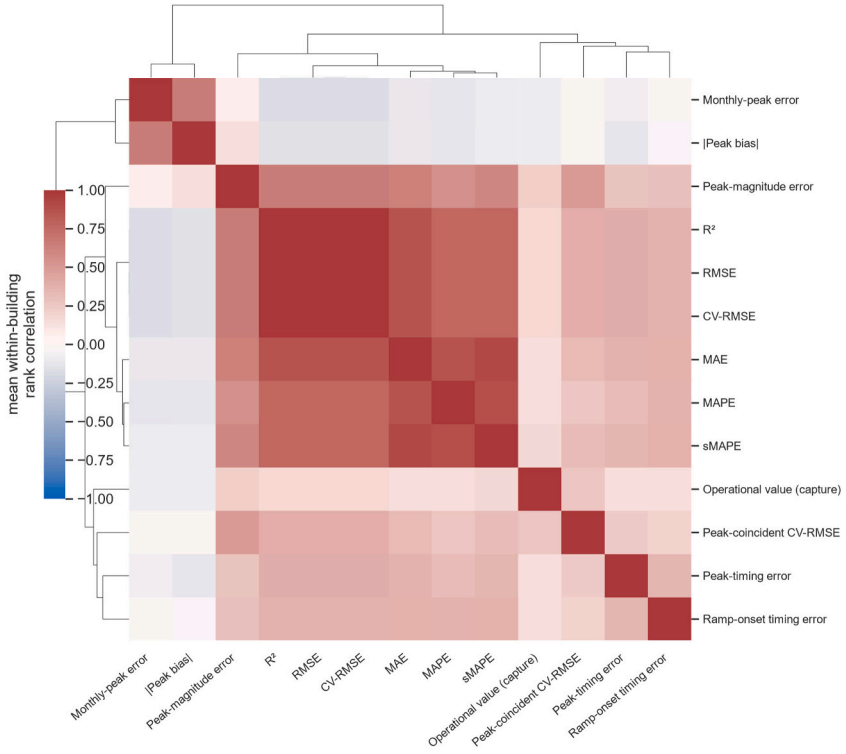


Fig. 7. Clustered map of mean within-building rank correlations among all metrics and operational value: capture sits with the peak metrics, apart from the average-error block.

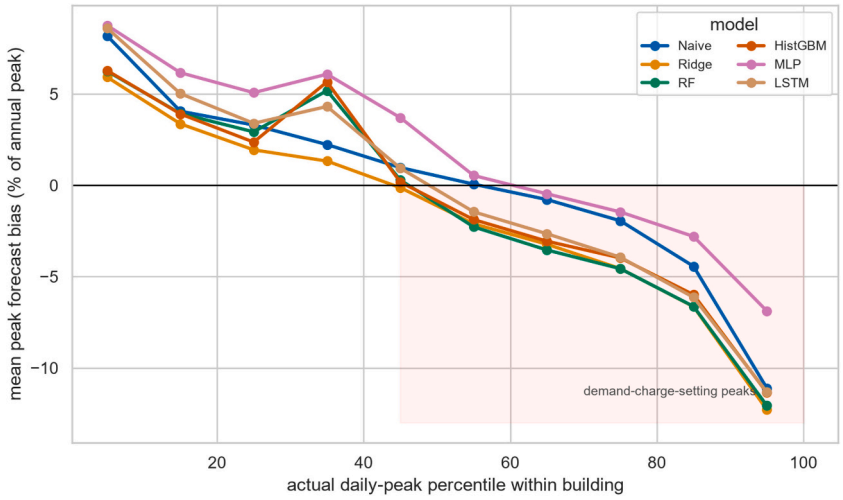


Fig. 8. Peak forecast bias conditioned on the day's peak percentile: small peaks are over-forecast, the demand-charge-setting peaks (shaded) under-forecast by 5–11% of the annual peak.

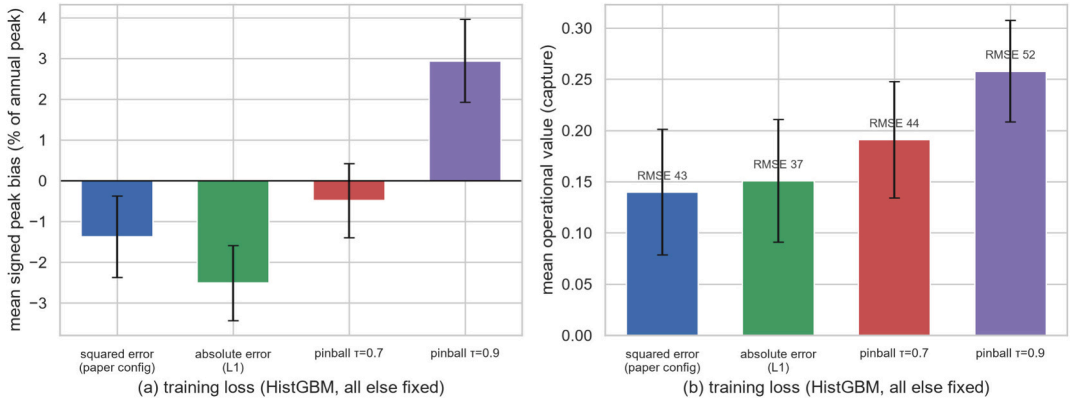


Fig. 9. The loss-function experiment (HistGBM retrained on all 200 buildings with only the loss changed). (a) Mean signed peak bias: symmetric losses (squared, absolute) under-forecast peaks; pinball losses reverse the bias. (b) Mean capture rises 84% from the squared-error to the $\tau = 0.9$ pinball variant while RMSE worsens (annotations).

Table 6
Sensitivity to tariff structure. The parenthetical value after the best value predictor is its mean within-building Spearman rank correlation with capture (as in Table 5). Statistics cover the 200 buildings, except under the TOU-only tariff, where 124 buildings have differentiated battery value (the six dispatches yield near-identical bills in the remainder).

Tariff	Demand charge	RMSE→value agreement (%)	Best value predictor (mean ρ)	Median / mean $ regret per building-yr
TOU energy only	none	16.9	MAPE (0.29)	$1 / $795
Flat energy + DC	$18/kW	15.0	RMSE (0.22)	$59 / $378
TOU + DC (primary)	$18/kW	12.0	peak-coincident (0.27)	$181 / $2872
Demand-heavy	$30/kW	11.5	peak-coincident (0.27)	$343 / $5505

demand charge the value gap is negligible (median regret $1): all models capture energy arbitrage to a similar degree, so forecast quality affects peak- and demand-related decisions but has little effect on energy shifting alone—and, consistently, the best value correlate there is a conventional percentage-error metric (MAPE, $\rho = 0.29$) rather than any peak metric. This tariff dependence is corroborated by a recent single-building study that found storage MPC economics robust to load-forecast error without demand charges but highly sensitive with them [36]. Across battery sizes (15–40% of peak power, 1–4 h energy) the RMSE–value agreement remains within a narrow 13–25% band (Fig. 10), indicating that the gap is not an artefact of battery sizing.

5.7. *Robustness II: Oracle horizon, controller variants, and battery degradation*

Oracle information horizon. The perfect-forecast benchmark in (8) uses the same receding 24 h horizon as the forecast-driven controller, which makes it the correct like-for-like reference for isolating forecast quality, but it is not the optimum over the billing period. *Re*-solving every building with a full-month perfect-foresight LP (§4.5) shows that the receding daily benchmark attains 85.5% of the monthly optimum's savings in aggregate under the primary tariff (per-building median gap 5.1% of achievable savings, 90th percentile 21.7%; under the demand-heavy tariff 89.2% and 2.2%), and a 48 h receding variant closes about a quarter of the remaining gap. Re-basing the capture denominator to the monthly optimum changes the mean captures only slightly (e.g., HistGBM 0.140 → 0.150; individual positive captures shrink with the larger denominator while the negative tails shrink in magnitude, so the means move little—Supplementary Table S4) and—because the



Fig. 10. Robustness to battery sizing across five configurations (15–40% of peak power, 1–4 h): the within-building Spearman(RMSE, value) (left axis, green) and the RMSE-best = value-best agreement (right axis, orange) stay in a narrow band. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

denominator is a per-building constant—leaves the within-building rankings, and therefore the agreement (12.0%), the mean Spearman (0.170), and every regret statistic, exactly unchanged. The same holds under the demand-heavy tariff (11.5%, $\rho = 0.109$).

Terminal state of charge and intra-day re-optimization. Constraining each daily LP to return the battery to its start-of-day state of charge changes the perfect-forecast benchmark bill by a negligible +$42/yr on average, and for the forecast-driven controllers raises bills modestly (median + $261/yr)—the midnight terminal condition truncates the overnight cheap-charging window and competes with deep shaving on peak days—but the headline comparison is unchanged (agreement 12.0%, mean Spearman 0.142). *Re*-optimizing at midday with the realized state of charge and running peak (a two-stage intra-day MPC, evaluated on the 60-building subset) improves bills for 40% of building-model pairs and leaves the accuracy-value gap in place (agreement 5.0% versus 8.3% for the base controller on the same subset; mean Spearman 0.236 versus 0.167). Neither controller refinement restores the alignment between accuracy and value.

Throughput, cycling and degradation. The dispatch cycles the battery heavily—210–216 equivalent full cycles per model-year on average (perfect-forecast benchmark: 218)—so we tested whether accounting for cycling cost changes the conclusions, in two ways. First, ex post: charging each model $0.01–0.10 per kWh discharged (bracketing the marginal degradation costs of ~$0.007–0.013/kWh estimated in the dispatch literature [75], and far beyond) moves the agreement only between 12.0% and 12.5%, the mean Spearman between 0.169 and 0.183, and raises the mean regret from $2872 to at most $3413; the models' throughputs are so similar (within ±4%) that degradation shifts all models' values nearly in parallel: the within-building rank statistics are essentially unchanged, although at the largest cycling costs the low-throughput seasonal-naïve baseline rises in the *population-mean* net-value ordering (Supplementary S5c). Second, degradation-aware dispatch: adding a $0.05/kWh discharge cost inside the LP objective (7) changes mean bills by −$13/yr and mean cycling from 212 to 211 EFC—the TOU arbitrage margin exceeds this cycling cost, so the dispatch is essentially unchanged. Battery degradation therefore affects the absolute economics of storage [76–78] but not the accuracy-versus-value comparison studied here.

5.8. Robustness III: Weather-forecast noise, random seeds, hyperparameters, and the model set

Weather-forecast noise. The main analysis assumes a perfect day-ahead weather forecast (§4.1). To test this assumption we corrupted the test-year (2017) weather inputs—training-year weather stays clean—with zero-mean Gaussian noise of realistic day-ahead magnitude: $\sigma = 2.0$ °C for air and dew-point temperature, 1.0 m/s for wind speed (clipped at zero) and 1.5 °C for the daily-mean temperature feature, i.i.d. across hours in the paper's original protocol. For the revision we repeated this test on the stratified 60-building subset with five independent noise realizations, and added five realizations of a temporally correlated AR(1) noise (autocorrelation 0.8 at 1 h, matched marginal σ, daily-mean feature recomputed from the noisy temperatures)—closer to real numerical-weather-prediction error—evaluating all realizations against the same trained models. Across the ten noise realizations the effects are minimal and stable: the population mean RMSE rises by less than 0.5% relative to the clean-weather control (the autoregressive load lags carry most of the predictive signal), the RMSE-best = value-best agreement varies between 10.0% and 20.0% around the clean-control value of 10.0% on this subset (six-model chance level 16.7%), and the mean within-building Spearman(RMSE, value) varies between 0.154 and 0.216 around the clean 0.163, with the temporally correlated realizations indistinguishable from the i.i.d. ones (Supplementary Table S10). The accuracy-value gap is therefore not an artefact of the perfect-weather assumption, and single-realization sampling error does not affect the conclusion.

Random seeds and hyperparameters. All stochastic models (RF, HistGBM, MLP, LSTM) were retrained on the 60-building subset under three random seeds and under an alternative hyperparameter configuration (RF 600 trees; HistGBM learning rate 0.10; MLP single hidden layer (64); LSTM hidden size 32). Across the three seeds the RMSE-best = value-best agreement is 10.0–15.0% (six-model chance level 16.7%), the mean within-building Spearman(RMSE, value) is 0.10–0.16, and the median regret of RMSE-based selection is $138–$151 per building-year; under the alternative hyperparameter configuration the corresponding values are 20.0%, 0.16 and $157 (Supplementary Table S9). The subset *mean* regret varies more across replicates ($3210–$18,190) because it is dominated by a few large buildings (§5.3); the rank-based headline statistics and the median regret are stable. No seed or configuration brings accuracy and value into alignment.

Composition of the model set. Recomputing the headline statistics with each model excluded in turn (five-model sets, chance level 20%) keeps the agreement at 15.5–18.0%—at or below chance in every case. Dropping the seasonal-naïve baseline reduces the mean within-building Spearman from 0.17 to 0.04, confirming that even the weak positive accuracy-value association is largely carried by the one model that is poor at both, not by meaningful discrimination among the trained models.

5.9. Portfolio-level evaluation

The main analysis operates each building independently. A campus or portfolio operator would instead aggregate: summing the member buildings' loads (and their forecasts) and dispatching one shared battery sized by the same rule against the aggregate peak. We repeated the full evaluation at the site level for the 12 BDG2 sites in our sample (1–38 sampled buildings per site), summing per-building forecasts into portfolio forecasts. Aggregation cancels forecast error exactly as the scaling literature predicts [79–81]: the median portfolio CV-RMSE falls to 7–11% (from 15 to 22% at building level), and the realized value rises—median capture 0.25–0.40 across models on the nine sites with at least ten buildings, versus mean captures of 0.05–0.14 at building level. The accuracy-value decoupling, however, persists: across those nine portfolios the RMSE-best model is the value-best in one, and the mean within-portfolio Spearman(RMSE, value) is +0.21. Two caveats follow. First, the two smallest "portfolios" (one and four buildings) show that an aggregate battery sized at 25% of a flat aggregate peak can lose money under forecast error—portfolio-level dispatch is not automatically safe. Second, coordinated but building-separate storage, portfolio-internal netting rules, and shared-storage placement [82] are richer settings than the simple aggregation studied here; our claim is only that moving from buildings to portfolios does not restore the alignment between RMSE and value.

5.10. Toward ex-ante, value-based model selection

The evaluation so far identifies the value-best model ex post. An operator must choose ex ante. To test implementability, we split the test year: all selection statistics are computed on January–June 2017 (H1) only—information an operator would actually possess—and the chosen model is deployed and scored on July–December (H2). Four ex-ante criteria are compared: H1 RMSE, H1 MAE, H1 peak-coincident CV-RMSE (the trailing-window version of metric (5)), and H1 simulated dispatch capture; benchmarks are the ex-post H2-best (upper bound), random selection, and the full-year-RMSE pick of the preceding sections. On the 194 buildings with non-trivial H2 savings: selecting by H1 simulated value attains the best median H2 outcome (capture 0.285; median regret $79/half-year) and H1 peak-coincident selection the best top-one agreement (20.1%), both ahead of full-year RMSE selection (0.270; $121; 13.9%)—but *no* criterion reliably identifies the H2 value-best model (agreement ≈ 20% at best, against an ex-post ceiling of 100%), because the trained models' values are closely separated and noisy across half-years. Three practical conclusions follow. (i) Value-based and peak-based selection are implementable ex ante—the required statistics need only a trailing window of realized data—and are modestly better than RMSE-based selection. (ii) Model *selection* among conventionally trained models recovers only a fraction of the ex-post value differences; the larger, more reliable lever is changing the training loss (§5.5), which raised value by 84% without any selection. (iii) For benchmarking—the primary concern of this paper—the ex-post analysis stands on its own: it measures what the accuracy leaderboard would have cost, whatever selection rule an operator uses.

5.11. Pre-cooling comfort case: A forecast-driven energy cost not captured by RMSE

Table 7 quantifies the pre-cooling case on the 181-building warm-climate subset (base zone). The robust effect is on cooling energy: the over-forecasting MLP spends the most seasonal cooling energy—in the base zone 4876 kWh versus 4810 kWh for the perfect-forecast controller (+1.4%), and the most of the six models in *all eight* zone configurations tested—because over-stating on-peak severity leads the controller to pre-cool more than necessary. Comfort violations, by contrast, are small in absolute terms (well under one degree-hour per cooling season in the base zone; near zero under a 26 °C comfort band or partial curtailment; at most ~5 degree-hours in the least favourable low-thermal-mass configuration) and, where they occur, track a model's skill at *ranking* which days are severe rather than the sign of its average bias: the seasonal-naïve model, unable to anticipate a hotter-than-last-week day, is the worst-comfort model (0.68 degree-hours versus 0.33–0.44 for the trained models in the base zone). The operational implication is consistent with the battery results: forecast error appears as an HVAC energy cost (and, at the margin, a comfort cost) that two models with the same RMSE can incur to substantially different degrees and that RMSE does not capture. We caution—as reflected in the framing of §4.6—that the magnitudes are zone-dependent and the controller stylized; we therefore treat the comfort case as an illustrative mechanism, not a population estimate of discomfort, and the battery dispatch results carry the paper's quantitative conclusions.

Table 7
Pre-cooling case, base zone, means over 181 warm-climate buildings and the May–September season. Full results for all eight zone configurations are in Supplementary Section S6.

Controller driven by	Cooling energy (kWh)	of which pre-cooling (kWh)	vs. perfect forecast (%)	Comfort violation (°C·h)	Violation hours
Perfect forecast	4810	2808	–	0.61	1.9
Naïve	4812	2775	+0.05	0.68	2.0
Ridge	4828	2854	+0.38	0.33	0.9
RF	4824	2848	+0.30	0.41	1.1
HistGBM	4822	2860	+0.25	0.44	1.1
MLP	4876	2940	+1.38	0.41	1.2
LSTM	4829	2881	+0.41	0.35	1.0

6. Discussion

Accuracy and value are only weakly coupled. A lower average error is, on balance, beneficial (the RMSE–value correlation is positive, 0.17, and reliably so; §5.3), but it is a weak and unreliable guide to operation: it identifies the value-best model no better than chance and is negatively associated with value in approximately one-third of buildings—and among the trained models alone the association essentially vanishes ($\rho \approx 0.04$). This appears to be structural rather than incidental. Operational value here is created at the peaks—the hours that set demand charges and that a battery or pre-cooling event must anticipate—while RMSE averages over all 8760 h. Formally, the dispatch objective (7) couples the bill to the forecast almost entirely through the demand-charge term and the on-peak energy window, i.e., through the error in perhaps 400–500 h per year; the remaining ~95% of hours dominate RMSE but are nearly irrelevant to the decision. Peak-coincident error is more informative precisely because it conditions on the decision-relevant hours ($\rho = 0.274$ vs 0.170)—and §5.10 shows a trailing-window version of it can be used ex ante—yet even it is weak in absolute terms, because value also depends on error structure (sign, timing within the peak window, persistence) that no scalar accuracy statistic summarizes. That is the substantive case for simulating the decision, not just scoring the forecast.

The loss function is a causal driver, verified experimentally. The loss-function experiment (§5.5) turns the regression-to-the-mean explanation from a plausible account into a demonstrated cause within the model class tested: holding the model class, features and hyperparameters fixed, symmetric losses (squared *and* absolute) produce the peak under-forecast, and asymmetric pinball losses reverse it and nearly double the captured value while worsening RMSE. This closes the loop with the value-oriented forecasting literature [31,35,47]: the divergence documented at scale here is exactly the headroom that loss-modification and decision-focused training exploit.

Implication for benchmarking. Ranking models by RMSE, MAE or R^2 optimizes a criterion that is, in these results, only loosely aligned with

operational benefit and in some settings negatively aligned with it. We recommend that building-forecast studies report at least one decision-relevant metric—peak-coincident error or, better, a simulated value such as demand-charge capture—alongside statistical accuracy, and that benchmarks which rank models do so on value when an application is named. The protocol is agnostic to how the forecast is produced: it applies unchanged to transfer-learning approaches and to the emerging zero-shot time-series foundation models [62,65,66], which we regard as a natural setting for future decision-relevant evaluation.

Implication for practice. Because operational value depends on accurate peak prediction (and tolerates a small positive bias), practitioners can (i) select the operating model by simulated value on a trailing window rather than by RMSE (§5.10); (ii) prefer peak-coincident error when a full simulation is impractical; or (iii) train with peak-weighted or asymmetric losses [34,40]—the option with the largest measured payoff here: retraining with a $\tau = 0.9$ pinball loss raised captured value by 84% on average (median $351, mean $4578 per building-year) with no change to the model class or features (§5.5). Each is a modest change that, based on these results, corresponds to up to several thousand dollars per building per year under demand-charge tariffs.

Scope of the contribution. This study is deliberately confined to the evaluation problem: we keep every model trained under conventional objectives and ask how its statistical accuracy relates to operational value. The loss-function experiment of §5.5 is a controlled manipulation used as evidence for the mechanism, not a proposal of a new training method; a full treatment of decision-focused or value-based training for building loads [44,47] raises separate estimation and optimization questions that lie beyond the present scope. Equally important is the scope of the *conclusion*: the accuracy-value gap we document is a property of peak-sensitive, asymmetric, demand-charge-driven decision settings. Where the downstream decision is smooth and symmetric in the forecast error—pure TOU energy arbitrage is the case in point—the models' realized values converge, so although the top-one agreement remains near chance (Table 6), the economic consequence of choosing by an accuracy metric becomes negligible (median regret $1); conventional accuracy metrics are then an adequate selection basis, and nothing in this paper argues against them there. The finding should be read as "RMSE is an incomplete criterion wherever peaks carry the money", not as "RMSE is generally misleading".

Threats to validity. We apply common tariffs and a single battery and thermal configuration across heterogeneous buildings, deliberately, to isolate the effect of forecast quality; absolute savings will differ with local tariffs and assets, but the central comparison is within each building across models and is invariant to these choices (§5.6–§5.7). The perfect day-ahead weather assumption is standard and is shown (§5.8) not to drive the result. The comfort study uses a representative zone rather than measured indoor temperatures, which BDG2 does not provide; we therefore present it as an illustrative mechanism (§5.11). Models are trained per building with fixed seeds and fixed, untuned hyperparameters; §5.8 shows the headline statistics are stable across seeds and configuration changes, and our claim concerns the structural divergence between accuracy and value rankings, not any model being optimally configured. Our tariffs follow North American commercial rate structures; in markets where demand charges are small or absent and capacity is priced differently, as in many European tariffs, the gap will be correspondingly smaller (Table 6 quantifies exactly this gradient). Demand charges are, in practice, frequently settled on 15 min maximum demand rather than hourly averages [48]; BDG2 provides hourly meters, so our simulation understates the true sub-hourly peaks and the associated demand-charge exposure for *all* models alike. Sub-hourly metering is known to raise measured peaks and change battery peak-shaving economics [83,84]; because sharper peaks make dispatch value even more dependent on the extreme hours that symmetric training attenuates, we expect finer resolution to widen rather than narrow the accuracy-value gap, but verifying this requires sub-hourly data and remains future work. Portfolio aggregation (§5.9) raises capture and shrinks relative errors but does not restore accuracy-value alignment; richer coordination schemes remain open. Finally, on statistical inference: the two headline statistics now carry explicit null comparisons (the empirical-independence permutation test for the agreement rate and the exact six-item null for the Spearman mean; §5.3), but we report nominal confidence intervals without correcting for the many metrics, tariffs and configurations examined, and we accordingly rest the conclusions on the reproducibility of the central findings across tariffs, battery sizes, controller variants, degradation assumptions, noise realizations, seeds and hyperparameters (§5.6–§5.8) rather than on any single significance threshold.

7. Conclusion

Across 200 buildings and a full out-of-sample year, the building load forecast with the lowest RMSE is the most operationally valuable one only 12% of the time—no better than chance—and choosing a model by RMSE forfeits a median of $181 per building per year (mean $2872, concentrated in the largest buildings) under a TOU-plus-demand-charge tariff, rising with the weight of the demand charge. RMSE explains little of the variation in realized value ($\rho = 0.17$, near zero among trained models), and peak-coincident error is a modestly stronger—though still weak—correlate ($\rho = 0.274$). A controlled experiment identifies the training loss as a causal driver of the divergence: symmetric error minimization systematically under-forecasts the demand-charge-setting peaks, and retraining the same model with an asymmetric pinball loss reverses the bias and raises captured value by 84% while worsening RMSE. These findings are robust to tariff structure, battery sizing, oracle horizon, terminal-state and intra-day controller variants, battery degradation, weather-forecast noise, random seeds, hyperparameters and the composition of the model set, and persist at the portfolio level. They are specific to peak-sensitive, demand-charge-driven applications—under pure energy arbitrage the models' values converge and the economic consequence of accuracy–value misalignment becomes negligible. We conclude that for such applications statistical accuracy is a necessary but insufficient criterion for evaluating building load forecasts, and we recommend—and provide a framework and reproducible protocol for—decision-relevant, value-based evaluation alongside it.

CRediT authorship contribution statement

ByungKi Jeon: Writing – review & editing, Writing – original draft, Conceptualization, Data curation, Methodology, Software. **Min-Hwi Kim:** Conceptualization, Methodology, Supervision, Validation, Writing – review & editing. **Jong-Kyu Kim:** Conceptualization, Methodology, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The 200 sampled buildings

The table below lists the BDG2 building identifiers of the stratified sample (site_use-type_name encoding as in BDG2), grouped by site.

Site (n buildings)	Building identifiers
Bear (15)	Bear_education_Austin, Bear_education_Bonita, Bear_education_Chad, Bear_education_Chana, Bear_education_Herb, Bear_education_Maryjane, Bear_education_Pattie, Bear_education_Sandy, Bear_education_Shelley, Bear_education_Yvette, Bear_education_Zandra, Bear_lodging_Erick, Bear_public_Orville, Bear_science_Alison, Bear_utility_Sidney
Bull (22)	Bull_assembly_Beau, Bull_assembly_Dorethea, Bull_assembly_Goldie, Bull_assembly_Vanessa, Bull_education_Dottie, Bull_education_Elva, Bull_education_Genie, Bull_education_Krista, Bull_education_Linnie, Bull_education_Luke, Bull_education_Magaret, Bull_education_Mervin, Bull_education_Myra, Bull_education_Olive, Bull_education_Racheal, Bull_lodging_Allen, Bull_lodging_Jeremiah, Bull_office_Ivette, Bull_office_Lilla, Bull_office_Myron, Bull_public_Hyun, Bull_services_Nadine
Eagle (20)	Eagle_assembly_Candice, Eagle_assembly_Ian, Eagle_education_Brianne, Eagle_education_Lino, Eagle_education_Luther, Eagle_education_Peter, Eagle_education_Roman, Eagle_education_Shana, Eagle_education_Teresa, Eagle_food_Kay, Eagle_health_Jodi, Eagle_lodging_Stephanie, Eagle_office_Bridgett, Eagle_office_Donovan, Eagle_office_Jeff, Eagle_office_Lane, Eagle_office_Lillian, Eagle_office_Mable, Eagle_office_Sheree, Eagle_office_Sonya
Fox (24)	Fox_assembly_Christie, Fox_assembly_Emma, Fox_assembly_Jerrod, Fox_assembly_Lakeisha, Fox_assembly_Terrell, Fox_assembly_Tony, Fox_education_Andre, Fox_education_Leona, Fox_education_Lesley, Fox_education_Melvin, Fox_education_Miguelina, Fox_education_Stacia, Fox_education_Theodore, Fox_food_Scott, Fox_lodging_Helen, Fox_office_Rowena, Fox_office_Susanne, Fox_office_Vicki, Fox_parking_Lynelle, Fox_public_Belle, Fox_public_Denny, Fox_religion_Maurice, Fox_retail_Manie, Fox_utility_Marian
Gator (4)	Gator_assembly_Marjorie, Gator_public_Nettie, Gator_public_Noe, Gator_warehouse_Constance
Hog (32)	Hog_education_Caridad, Hog_education_George, Hog_education_Jordan, Hog_education_Madge, Hog_education_Robert, Hog_food_Morgan, Hog_health_Hisako, Hog_industrial_Jay, Hog_industrial_Jeremy, Hog_industrial_Joanne, Hog_lodging_Nikki, Hog_lodging_Retha, Hog_office_Almeda, Hog_office_Byron, Hog_office_Cornell, Hog_office_Elke, Hog_office_Garrett, Hog_office_Gustavo, Hog_office_Man, Hog_office_Mike, Hog_office_Napoleon, Hog_office_Randi, Hog_office_Simon, Hog_other_Lynette, Hog_other_Noma, Hog_parking_Antoinette, Hog_parking_Bernardo, Hog_public_Crystal, Hog_science_Max, Hog_services_Joe, Hog_services_Kerrie, Hog_warehouse_Louise
Lamb (13)	Lamb_assembly_Delilah, Lamb_assembly_Elinor, Lamb_assembly_Ossie, Lamb_assembly_Tasha, Lamb_education_Antonio, Lamb_education_Augustine, Lamb_education_Daniel, Lamb_education_Gabrielle, Lamb_education_Mae, Lamb_education_Rick, Lamb_education_Robin, Lamb_education_Rodrigo, Lamb_public_Vania
Moose (1)	Moose_education_Lori
Peacock (11)	Peacock_education_Pasquale, Peacock_education_Patience, Peacock_education_Shelly, Peacock_education_Yolanda, Peacock_lodging_Nova, Peacock_lodging_Sergio, Peacock_lodging_Wes, Peacock_office_Burl, Peacock_office_Effie, Peacock_office_Elton, Peacock_office_Norman
Rat (38)	Rat_assembly_Cristina, Rat_assembly_Jannie, Rat_assembly_Mirta, Rat_assembly_Rosemarie, Rat_assembly_Teodoro, Rat_assembly_Trudy, Rat_education_Barney, Rat_education_Dreama, Rat_education_Guillermo, Rat_education_Irma, Rat_education_Jacob, Rat_education_Jesse, Rat_education_Marianna, Rat_education_Meghan, Rat_education_Sina, Rat_education_Vicky, Rat_education_Yu, Rat_health_Rosaria, Rat_office_Adele, Rat_office_Jeannie, Rat_office_Loyd, Rat_other_Lan, Rat_public_Berry, Rat_public_Dexter, Rat_public_Faye, Rat_public_Frederick, Rat_public_Joann, Rat_public_Johnny, Rat_public_Kermit, Rat_public_Kimber, Rat_public_Laverne, Rat_public_Margart, Rat_public_Mark, Rat_public_Neil, Rat_public_Ramon, Rat_public_Tamra, Rat_religion_Kathy, Rat_warehouse_Doretta
Robin (14)	Robin_education_Della, Robin_education_Julius, Robin_education_Kiera, Robin_education_Leslie, Robin_education_Madeline, Robin_education_So, Robin_education_Terrance, Robin_education_Zenia, Robin_lodging_Elmer, Robin_lodging_Janie, Robin_lodging_Oliva, Robin_office_Adolph, Robin_office_Antonina, Robin_office_Gayle
Wolf (6)	Wolf_education_Anisa, Wolf_education_Bobby, Wolf_office_Joana, Wolf_retail_Harriett, Wolf_retail_Toshia, Wolf_science_Alfreda

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.enbuild.2026.118339.

Data availability

Data will be made available on request.

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